

AI Engineering (Week 1)

aie26

Topic 02 : Motivating Example

Part 02 : Introduction to Data Operations

Preparation

Data Handling

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Building Models

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Outline

- Characteristics of data sets
- Operations on tabular data

Wrap up

AI Engineering (Week 2)

Introduction



Motivating Example

Preparation

Data Handling

Exploring Data 1

Exploring Data 2

Building Models

Prediction

Regression
1

Classification
1

Regression
2

Classification
2

Clustering

Wrap up

Outline

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Data sources

Type	Format	Example	DBMS	Language	Readiness for ML
Relational	Table	Transactions	MySQL, Postgresql, ...	SQL	Maps to dataframe
Flat	Key + Value	Caches	Redis, mem- cached, ...	DBMS-Specific	Not rich enough
Document	Serialised objects	Tweets	Mongodb, Cassandra. ...	MQL, CQL	Too rich
Graph	Nodes and edges	Social rela- tionships	Neo4j, Dgraph, ...	Gremlin, Cypher, DQL, ...	Specialised analyses
Columnar	DataSet	Logs	HBase, Spark DataSet	Hive QL, Spark SQL	Maps to dataframe

Generally, **rich** flat data representations are best suited to machine learning

Preparing data

➤ Data Preparation is the first step in data mining

In practice, data can be

- structured or unstructured,
- consolidated or scattered,
- consistent or inconsistent,
- clean or with error.

➤ ML prefers structured, consolidated, consistent data, as clean as possible.

The `auto_mpg.csv` dataset already has these characteristics.

The auto-mpg dataset

1	mpg	cylinders	displacement	horsepower	weight	acceleration	model year	origin	car name
2	18	8	307	130	3504	12	70	1	chevrolet chevelle malibu
3	15	8	350	165	3693	11.5	70	1	buick skylark 320
4	18	8	318	150	3436	11	70	1	plymouth satellite
5	16	8	304	150	3433	12	70	1	amc rebel sst
6	17	8	302	140	3449	10.5	70	1	ford torino
7	15	8	429	198	4341	10	70	1	ford galaxie 500
8	14	8	454	220	4354	9	70	1	chevrolet impala
9	14	8	440	215	4312	8.5	70	1	plymouth fury iii
10	14	8	455	225	4425	10	70	1	pontiac catalina
11	15	8	390	190	3850	8.5	70	1	amc ambassador dpl
12	15	8	383	170	3563	10	70	1	dodge challenger se

Notes

- ① The data is structured, consolidated, consistent and clean
- ② The first row contains the headings (column names), the remaining rows are the observations (cases, instances,...)
- ③ Each column stores a *variable*. Other terms include attributes and targets.
- ④ Machine learning uses combinations of these columns to build models.

Understanding the auto-mpg dataset: Column Sufficiency

Learning

- Given a collection of columns (a *projection* of the full dataset), how does this help the machine to learn?
- It provides example data representing a phenomenon. . .
- But what collection of columns to use?
- Depends on the problem we wish to solve. . .
- Can it be used to predict some quantity (a target)?
- And what does *prediction* mean?
- Are there other forms of learning apart from being able to predict?

Applied to auto-mpg

- Given explanatory variables displacement, horsepower, weight, can we predict *mpg* (target)?
- Are all these explanatory variables needed, or could some be dropped?
- Are additional explanatory variables needed, either from auto-mpg or elsewhere?
- How do we measure the *quality* of a prediction?
- What other learning can be derived from the chosen column collection?

Understanding the auto-mpg dataset: Row Sufficiency

Selection

- Do we have enough, too many or just enough observations?
- If we project the data, we might have multiple rows with the same explanatory values but different target values...
- ...Is this good or bad?
- How can we exclude unnecessary or incompatible observations?
- ... We can use *selection* (also known as *restriction*) - but how do we choose which rows to keep?

Example

```
SELECT displacement, horsepower, weight, mpg  
FROM auto_mpg  
WHERE horsepower > 79;
```

- SELECT clause: projection (restricting columns: column sufficiency)
- WHERE clause: selection (restricting rows: row sufficiency)

Understanding the auto-mpg dataset: Summarising

Often in data mining, we need to “see the wood for the trees”

- Generally, we want our data to be as granular as possible - more detail is *better*
- ... We can remove detail if needed, but cannot add it later
- Is duplicate data good or bad in machine learning?
- ... GOOD: estimating variability in a quantity, using statistical methods
- ... BAD: can obscure useful implementation - an aggregated value might be more useful
- Can generate summaries in three ways
 - 1 Sampling - reduce/remove row duplication
 - 2 Banding - reduce the cardinality of a column
 - 3 Grouped Aggregation - roll up by level, aggregating as needed

Understanding the auto-mpg dataset: Sampling

➤ To reduce (but not remove) duplication, sampling can be a good compromise

- To reduce bias, the sample should be random (each row has the same probability of being picked)
- Number of rows to keep in the sample is a compromise
- Can reduce runtime while allowing estimates of the uncertainty in a predictive model
- However, an aggregated column might be a better choice

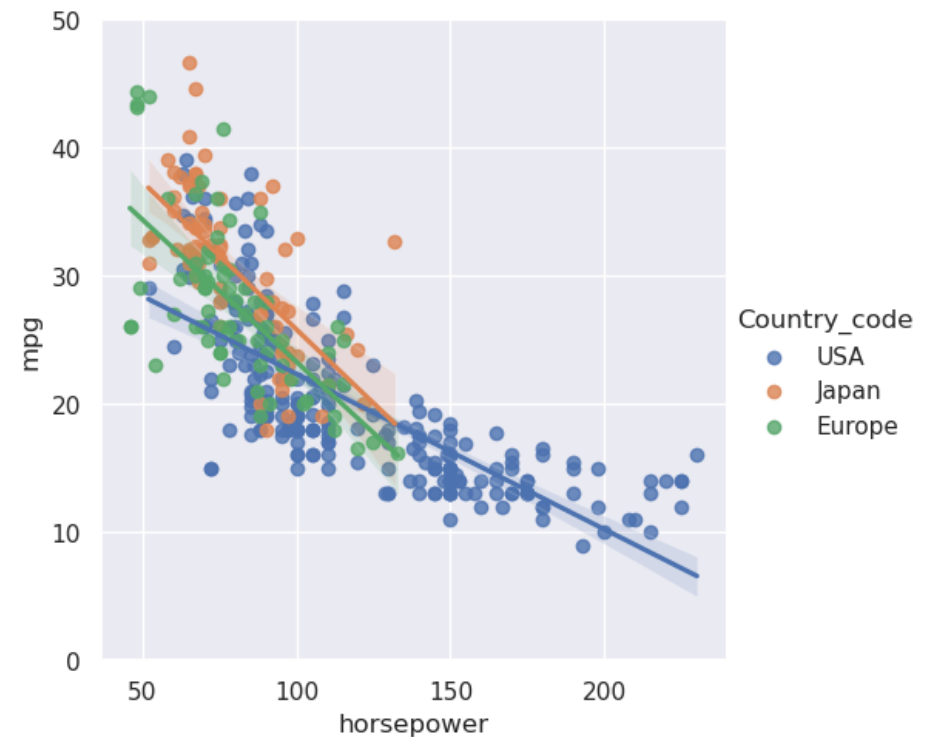
```
-- Return a random sample of 3 Japanese cars  
SELECT *  
FROM AutoMpg  
WHERE originID = 3  
ORDER BY RANDOM()  
LIMIT 3;
```

Understanding the auto-mpg dataset: Banding

A new column with reduced cardinality is often more understandable

- Sometimes a column with fewer distinct values offers fresh insights
- ...Derive carMaker from carName - compare different manufacturers
- When a column contains real numbers, currency, etc., this is very noticeable
- ...*Banding* - assigning those numbers to non-overlapping ranges can simplify analysis

```
SELECT substr(carName,1,instr(carName,' ')-1)
       AS carMaker
,CASE
  WHEN horsepower < 130 THEN 'low'
  ELSE 'high'
END AS horsepowerGroup
FROM AutoMpg;
```



Understanding the auto-mpg dataset: Grouped Aggregation

➤ A grouped aggregation changes the effective key structure

- Aggregations include: `MIN()`, `SUM()`, `COUNT(DISTINCT ...)`
- ... Take a set of values, compute an aggregate value
- Sets can be partitioned by grouping variable, aggregate applied to each partition
- ... Example: average mpg per country of manufacture

```
SELECT originID, AVG(mpg)
FROM AutoMpg
GROUP BY originID;
```

Summary

- Semantically rich, flat data is preferred for machine learning
- Ideally, this data would also be structured, consolidated, consistent and clean
- Several data operations were described, using the AutoMpg dataset as an example data source
 - Projection
 - Selection
 - Summarising: Sampling, Banding, Grouped Aggregation

➤ Your task is to apply this to datasets using the python toolchain. ➤